

SPECIAL REPORT

The New Rules of AI Power

How DC power is changing what
it takes to build and scale AI



September 2026

Preface

The modern power system was born at the end of the nineteenth century as cities were growing, factories were expanding, and electricity began to transform how people lived and worked. The new economy needed electricity for lighting, streetcars, elevators, motors, and industrial machinery. It also needed a practical way to generate electricity at scale and deliver it over long distances to where it was needed.

Alternating current (AC) made that possible. Its voltage could be raised for efficient transmission over long distances and then lowered again near homes, businesses, and factories to levels that were safe and practical to use. That advantage helped establish a model built around large, centralized power plants connected to customers through transmission and distribution lines. Over the 20th century, that architecture expanded into the grid we know today. It was extraordinarily effective for the mechanical age it was built to serve.

Today's digital age looks very different. The machines that relied on motors and mechanical systems are increasingly enabled by computers, artificial intelligence, and software. Factories run on automation and robotics. Hospitals depend on advanced imaging, monitoring, and diagnostic systems. Everyday life depends on smartphones and connected devices. Offices, communications networks, and data centers rely on computing at enormous scale. At the heart are semiconductor-based electronics powered by direct current (DC). Yet most of the electricity that powers the digital economy is still delivered as AC.

Until recently, that mismatch was manageable. Data centers brought AC power into the facility and converted it to DC at the rack level. At conventional rack densities, the amount of equipment, copper, and space required remained relatively modest. AI changes the scale. Rack power is expected to reach 600 kilowatts (kW) by 2027, with a path toward 1 megawatt (MW).¹ At those densities, converting and moving power becomes far more costly and complex.

That leaves data center developers with a consequential choice: continue with the existing AC power model and convert it to DC, or generate DC power onsite and avoid much of the conversion altogether. The decision reaches beyond electrical design, affecting equipment, construction timelines, reliability, and available space.

The financial stakes are enormous. In 2026, the five largest U.S. hyperscalers are expected to deploy about \$750 billion in capital.² At that scale, even relatively small differences in infrastructure cost translate into billions of dollars.

This report compares the leading approaches to delivering DC power at AI scale and the choices they create for developers. The path they choose will help determine whether power infrastructure enables the next generation of AI compute or becomes a constraint.

¹ NVIDIA published rack specifications and SemiAnalysis Datacenter Industry Model

² S&P Global Ratings, S&P Global Ratings' View On Artificial Intelligence And Hyperscalers, Estimate includes Alphabet, Amazon, Meta, Microsoft, and Oracle

Executive Summary

- 1. Rapidly growing compute density is driving a shift in data center electrical architecture.** AI compute is outrunning the power infrastructure built to support it. Power demand per rack has increased roughly twenty-fold since 2020 and is expected to reach 600 kW by 2027, with a path toward 1 MW.³ At those densities, scaling today's AC power architecture is more cumbersome and expensive: impractical amounts of copper, additional electrical equipment, and valuable space devoted to power infrastructure rather than compute. The industry's response is 800V DC, which can deliver far more power with less current and is emerging as the standard for next-generation AI systems. NVIDIA will require 800V DC beginning with its Rubin Ultra and Kyber rack architecture. For data center developers, the question now is how 800V DC power will be supplied.
- 2. The default assumption is to preserve existing AC power sources and convert their output to DC.** The grid and most sources of data center power still deliver AC power, even as next-generation AI systems move to DC. The industry bridges the gap to bring AC power into the data center and convert it to 800V DC. That still requires substantial electrical infrastructure with year-long lead times. Some designs also depend on technologies with less operating history, such as solid-state transformers that add to supply-chain and operational risk. The result is a new power architecture built on old infrastructure.
- 3. There is another path: generate DC power natively and avoid costly and complex AC-to-DC conversion.** If AI racks require DC power, the most direct path is to generate it from the beginning. This eliminates much of the equipment required to convert AC power and reduces the number of components to buy, install, and maintain. It also reduces dependence on long-lead electrical equipment and potential points of failure. With fewer conversion stages, less power is lost along the way and less space is consumed by electrical infrastructure. The result is a more streamlined and less costly path from generation to consumption.
- 4. Fuel cells are the only DC-native source that can provide continuous power at gigawatt scale.** Very few onsite power sources can meet all three requirements of an 800V DC data center: generate DC natively, run continuously, and scale to gigawatt loads. Solar generates DC but cannot provide continuous power. Batteries operate on DC but store electricity rather than generate it. Turbines and reciprocating engines can provide continuous power at scale, but they generate AC and retain much of the same conversion infrastructure as a grid-powered site. Fuel cells are the exception: they generate DC natively and can run continuously. Bloom Energy is the only fuel cell provider that operates at scale for large AI data centers and combines all three requirements into one onsite power source.

³ NVIDIA published rack specifications and SemiAnalysis Datacenter Industry Model.

- 5. Bloom's architectural advantage reduces data center non-compute CAPEX by 27%, and five-year total cost of ownership by 9%.** At gigawatt scale, what looks like an electrical design choice becomes an economic one and changes where billions of dollars are spent. For a 1 GW data center, minimizing upstream distribution and conversion infrastructure lowers the five-year total cost of ownership by approximately \$5.5 billion, or about 9% of total facility cost. Of that, roughly \$3.6 billion comes from a 27% reduction in non-compute CAPEX. That capital can instead go toward revenue-generating IT equipment, enough to fund roughly 8% more compute for the same overall investment. The advantage extends beyond savings. An 800V DC-ready facility can deploy each new generation of higher-density chips as it becomes available, shorten the path to deployment, and increase the economic value of the data center.

1. Rapidly growing compute density is driving a shift in data center electrical architecture

Rack power demand is increasing faster than the power infrastructure built to support it. In 2020, a typical rack drew about 10 kW. Today, that figure is closer to 200 kW, and by 2027 rack power is expected to reach 600 kW (Exhibit 1), equivalent to the electricity consumption of 500 US homes.⁴ The electrical infrastructure designed for today’s racks cannot support the next generation arriving next year.

EXHIBIT 1: Each generation of rack architecture packs more compute into a single rack, and power demand rises with it.

Rack power demand evolution

Timeline	Rack Architecture	GPU	Rack Solution	GPUs/ Rack	Total Power Consumption
2020	EIA	Ampere	DGX A100 SuperPod	8	~10 kW
2022	MGX	Hopper	GH200 NVL32	32	~40 kW
	Oberon	Blackwell	GB200 NVL36	36	~70 kW
		Blackwell	GB200 NVL72	72	~120 kW
2024		Blackwell Ultra	GB300 NVL72	72	~135-140 kW
2026		Rubin	Vera Rubin NVL72	72	~200 kW
2027		Rubin Ultra	Rubin Ultra NVL72	72	Not Published
2028	Kyber	Rubin Ultra	Kyber NVL144	144	~600 kW–1 MW
		Feynman	Solution Not Yet Disclosed		

Source: NVIDIA, “800 VDC Architecture for Next-Generation AI Infrastructure”, SemiAnalysis

⁴ US Energy Information Administration, average annual electricity use per residential customer. At roughly 10,500 kWh a year per household, 600 kW of continuous draw is equivalent to about 500 homes.

Delivering that much power in lower-voltage power architecture causes electrical losses and requires significantly more equipment, copper, and space that could be used for compute. In addition, the required 400 pounds of copper for each rack becomes impractical.⁵

Therefore, the industry is converging on 800V DC as the power architecture for next-generation AI systems. Higher voltage allows the same amount of power to be delivered at lower current and requires significantly less copper and equipment. For data center developers, this shift is a necessity rather than a choice.

AI-native companies with these rack architectures will feel the impact first. Beginning with Rubin Ultra and Kyber rack architecture, NVIDIA has specified 800V DC for 2027 and all generations after it. Facilities not designed for the new architecture will require substantial changes. Developers planning for that generation of chips must build for 800V DC whether or not they have chosen the standard.

The transition is already underway. An August 2026 Bernstein survey of 50 North American data center procurement leaders found that 15% already have an 800V DC project under construction or in operation, and another 29% expect to begin one by 2027.⁶ Bloom Energy's 2026 Mid-Year Pulse Data Center Power Report found a similar trend where data center leaders expect DC-based architectures to account for 58% of new deployments by 2030.⁷

The question now is how 800V DC power will be supplied. Power architecture is determined years before a facility opens and is expensive to revise after the fact. With 800V DC adoption expected to accelerate in 2028 and 2029, developers targeting that window need to be making those decisions now.

⁵ NVIDIA, "NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories," NVIDIA Technical Blog, May 2025.

⁶ Bernstein survey of 50+ US data center operators, June 2026.

⁷ Bloom Energy, 2026 Data Center Power Report — Mid-Year Pulse, June 2026 (n=90 US data center developers).

2. The default assumption is to preserve existing AC power sources and convert their output to DC

The grid and most power sources provide AC power, even as next-generation AI systems move to DC. The prevailing response is to preserve that model: source AC power from the grid and convert it to 800V DC. Even when racks move to 800V DC, a grid-powered site still needs a substation, medium-voltage switchgear, AC-to-DC conversion equipment, and backup power to carry the load through an outage. The equipment is costly and leaves developers exposed to year-long lead times on large transformers and switchgear (Exhibit 2). As a result, developers incur expense and schedule risk associated with conventional AC architecture.

EXHIBIT 2: Key electrical equipment for grid-powered designs carries lead times measured in years.

Grid-powered electrical equipment lead times

Grid-powered 800V DC			3-4 years Potential wait time for long-lead equipment
Large power transformers	3-4 yrs		
Medium voltage switchgear, DC spec	2-3 yrs		
High voltage circuit breakers	125 wks		
Solid state transformer	No estimate available ¹		

¹ The solid-state transformer is still at pilot stage, supplied by a handful of vendors with no delivery track record

Sources: Environment+Energy Leader, "Grid equipment bottleneck isn't just a data center problem," July 2026, citing Reuters and Wood Mackenzie; Giga Energy, "The constraints that cause lead time delays for data centers," August 2026, citing Bloomberg; Terrapin Construction Group, "Switchgear, transformer and generator lead times in 2026," June 2026; Bernstein, Solid-State Transformers: 800V, Zero Windings, 19 August 2026

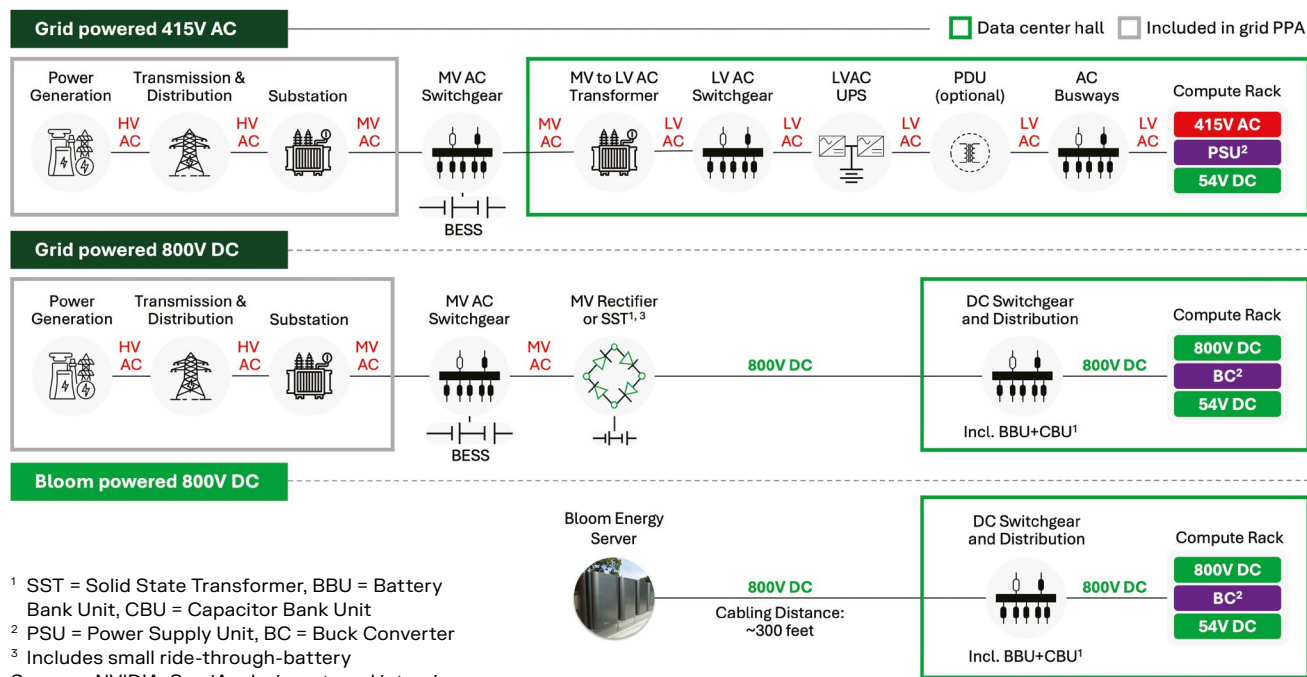
The industry's answer to multiple transformation and conversion steps is a solid-state transformer that collapses medium voltage transformation and AC-to-DC conversion into one system and replaces several pieces of conventional electrical equipment. But solid-state transformers are newer technology with less operating history. No suppliers are producing solid-state transformers at the volumes a gigawatt data center needs, and service practices and operating history are thinner than those for conventional transformers. Developers must choose between mature equipment that carries the full cost and lead-time burden, or new equipment that does not yet exist at scale. The underlying constraint remains the same: power arrives as AC.

3. There is another path: generate DC power natively and avoid costly and complex AC-to-DC conversion

Starting with DC eliminates much of the equipment required to convert AC power (Exhibit 3). Power produced as DC needs no conversion, simplifying the path from the generation source to the rack. There are fewer components to purchase, install, operate, and maintain. It's a more reliable system with fewer points of failure and less electrical maintenance across the life of the facility.⁸ Fewer conversion stages also reduce electrical losses and free up space inside the data center for compute.

EXHIBIT 3: Moving from grid AC to onsite DC generation removes equipment between the power source and compute rack.

Power path from source to compute rack, by architecture



Combining multiple DC sources is simpler than integrating multiple AC sources. A DC system keeps voltage steady, while AC systems must keep voltage, frequency, and timing aligned for every source at once, which requires additional equipment and controls. The practical effect is that DC capacity can be added as demand grows without reconfiguring equipment.

⁸ NVIDIA and OCP Global Summit materials, October 2025, which attribute to the 800V DC architecture up to 5% better efficiency, roughly 70% lower maintenance cost and up to 30% lower total cost of ownership.

4. Fuel cells are the only DC-native source that can provide continuous power at gigawatt scale

When power is generated as DC from the outset, data center sites do not need the conversion equipment described in Section 2. Equipment in an AC-powered conventional build-out is costly; eliminating it significantly lowers capital costs. The challenge is finding an onsite power source that combines three requirements: native DC generation, continuous output, and the ability to operate at data center scale. Most available technologies meet only some of these requirements (Exhibit 4).

Solar generates DC power natively, but its output is intermittent and cannot independently support a continuous gigawatt-scale load. Batteries also operate on DC, but they store rather than generate electricity, which limits their role as a primary source of power. Turbines and reciprocating engines can provide continuous power at scale, but they generate AC; therefore, an 800V DC data center powered by those technologies still requires the transformation and conversion infrastructure and encounters the long lead times for equipment described in Section 2.

Starting with DC also leaves the distribution architecture open. Bloom's fuel cells deliver 800V DC with a center point, allowing customers to use either a monopolar or $\pm 400V$ DC bipolar architecture from one platform. Fuel cells are the only DC-native generation technology capable of delivering continuous DC power at AI scale and Bloom is the only fuel cell provider currently delivering that capability at scale for AI data centers.

EXHIBIT 4: Fuel cells are the only source that generates DC natively, runs continuously, and scales to gigawatt loads.

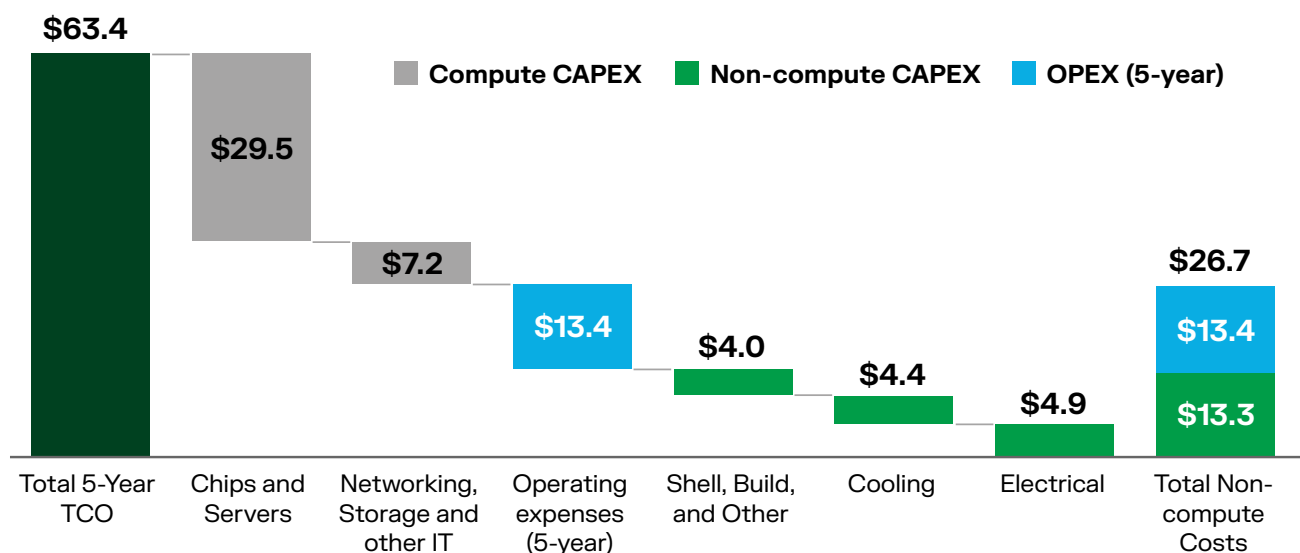
AC power sources	
	Grid (800V DC feed) Most of the equipment as regular AC, significant energy loss, high equipment cost, and supply chain issues
	Onsite turbines or recips Similar issues to the grid with high equipment cost, and supply chain issues
DC power sources	
	Solar High potential power output but intermittent; needs paired battery storage or other generation
	BESS Stores power but doesn't generate it; needs its own charging source
	Fuel cells Native 800V DC output, minimal losses from generation to rack, and reduced need for electrical equipment

5. Bloom's architectural advantage reduces non-compute CAPEX by 27% and five-year total cost of ownership by 9%

Chips, servers, and other IT costs account for more than half of a data center's total cost and are set by the chip roadmap, so a data center developer has little influence over them. Non-compute costs are about 40% and cover the electrical system, cooling, buildings, and five-year operating costs. Exhibit 5 shows the breakdown of costs for a conventional grid-powered AC data center scenario.

EXHIBIT 5: Non-compute costs represent ~40% of the 5-year TCO.

Estimated 5-year total cost of ownership for grid-powered AC, \$B

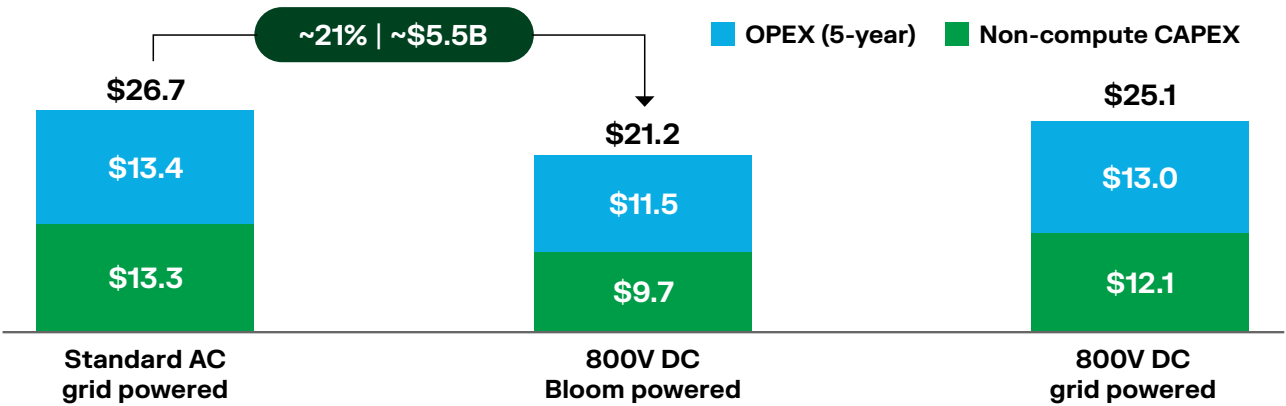


Budgets are under scrutiny and every dollar spent on buildings and electrical infrastructure is a dollar that can't go toward compute.

On a 1 GW build, a Bloom-powered, DC-native data center costs about \$5.5 billion less over five years (Exhibit 6), roughly 21% of five-year non-compute costs compared to a conventional AC build powered by the grid.

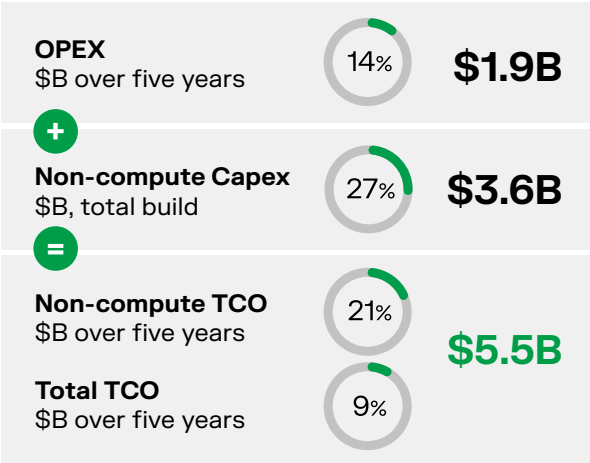
EXHIBIT 6: Bloom-delivered 800V DC cuts five-year non-compute costs by \$5.5 billion, or about 21% compared to grid-powered standard AC.

Non-compute total cost of ownership comparison, \$B



CAPEX = non-compute (electrical, cooling, shell/build). OPEX = 5-yr present value energy, software, maintenance, personnel, and other cost. Source: Bloom Energy Analysis, Expert interviews; SemiAnalysis, NVIDIA

Of that total, \$1.9 billion comes from lower operating costs and \$3.6 billion from lower capital cost, which result in a 14% reduction in OPEX and a 27% reduction in non-compute CAPEX.



Savings are driven by three changes to the build:

- **Overall operating costs**, driven by removing equipment that is expensive to service and maintain.
- **Non-compute capital costs for electrical** as key infrastructure components like substation, solid-state transformers, and battery systems are removed from the design.
- **The rest of non-compute capital costs** (e.g., shells and design) because a smaller electrical footprint reduces overall building size.

A design with fewer conversion stages loses less power and reduces energy costs. At gigawatt scale, those savings materially improve project economics. For example, a Bloom-powered 800V DC build with the same investment of \$49.9 billion could fund approximately 8% more compute than a conventional AC build with the \$3.6 billion in reduced CAPEX.

Almost everything else about a data center’s cost is inherited. The chip roadmap sets the IT spend, the site sets the land, and the market sets the price of construction. Electrical architecture is an exception and is the reason two facilities of identical capacity and running identical hardware can differ by billions of dollars in what they cost to build and operate.



Conclusion

The chip roadmap is increasingly setting the terms for data center design. The racks arriving in 2027 assume a power architecture that most facilities do not yet have. How developers supply that power will determine whether they can deploy each new generation as it arrives or retrofit their infrastructure to catch up. Starting with DC frees more capital for the part of the data center that creates the most value: compute.

Appendix: Model Basis

This appendix sets out the basis of the cost comparison. The comparison can be rebuilt on a different platform, scale, cost year or price set, provided compute is held constant and only the power source and electrical path vary.

Table A.1: Case definitions

Case	Standard AC (baseline)	800V DC from the grid	800V DC generated on site
Power source	Utility, alternating current	Utility, alternating current	Bloom fuel cells, direct current at source ¹
Power-IT (MW)	1,000	1,000	1,000
Rack / chip architecture	Vera Rubin NVL72	Vera Rubin NVL72	Vera Rubin NVL72

1. Assumes an average 300-foot run to the edge of the data center shell.

Table A.2: Basis of the comparison

Item	Basis
Facility	Greenfield data center sized to a target IT load.
Compute held constant	The same rack platform and rack count in all three cases, so demand is identical and the cases are comparable outside IT compute.
Rack platform	NVIDIA Vera Rubin NVL72, with 54V distribution inside the rack. Rack count follows from the platform and the IT load.
Energy demand	Power usage effectiveness converts IT load into gross facility load, which drives energy demand.
Grid power price	This assumes a data center specific large load tariff negotiated to include grid upgrades such as large-load interconnection, transmission, and step-down to medium voltage on top of typical commercial grid power prices.
On-site generation	The plant sits outside the developer's capital cost and is billed as a PPA, so it lands in operating cost.
Cost basis	2029 prices, US dollars, single site. Priced per rack for IT, per MW of IT load for everything else.
Cost measured	One-time capital cost plus five years of operating cost, escalated and discounted to present value.
Units	Quoted per MW of IT load, which keeps cases comparable at any scale. At 1,000 MW, millions per MW and billions for the whole build are the same number.
Difference between cases	The power source and the electrical path to the rack; every difference in the results traces to those choices.
Conserved across cases	Compute hardware, cooling technology, staffing, software licensing, utilization and site conditions.

Table A.3: Scope

Category	Treatment
Capital cost, IT/compute	GPU racks priced per rack, plus networking, storage and other IT built up per rack. Identical in all cases.
Capital cost, non-IT	Priced per MW of IT load: electrical equipment from boundary to rack, cooling, shell and build, project management and engineering, lighting, fire protection, security, enclosures and raised floor.
Operating cost	Energy as a PPA based on facility demand, infrastructure maintenance, software licenses, personnel, other site operating cost, property insurance and tax.
Excluded	Land, interconnection charges and equipment up to medium voltage distribution (included in PPA), financing, tax and depreciation, decommissioning and residual value, IT refresh, and any value for earlier first power.

Limitations to carry forward

- These are modeled figures, with no site-specific engineering or vendor bids behind them. Some priced equipment is not yet in volume production, so its specifications are vendor targets, not measured performance.
- Cooling and building unit costs are assumed to scale linearly with IT load.
- No value is placed on earlier first power or on the reliability gain from fewer conversion stages; both favor the on-site DC case.



About Bloom Energy

Bloom Energy empowers enterprises to meet soaring energy demands and responsibly take charge of their power needs. The company's fuel cell systems provide ultra-resilient, clean and highly scalable onsite electricity for Fortune 500 customers around the world, including data centers, semiconductor manufacturing, large utilities and other commercial and industrial sectors, as well as mission-critical organizations in local communities such as hospitals, college campuses and retailers. Headquartered in Silicon Valley, Bloom Energy employs more than 2,000 people worldwide and manufactures its systems in the United States. For more information, visit bloomenergy.com/industries/data-center-power

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